Lahore University of
Management Sciences

Reversing the Slide: Understading and Mitigating In-Person Class Absenteeism at LUMS

10 April 2026

Prepared by
LUMS Learning Institute

From Assumptions to Evidence

Focus: Patterns, Drivers, Possible Interventions, and Recommendations

Analysis : Survey Data of approx 400 LUMS students , TA Data Compilation and Instructor Data evaluation

Pilot study was conducted prior to the actual deployment
-Survey was conducted using google forms and was emailed to LUMS students
Time frame: December 2025- January 2026

Starting point: Evidence from Literature

Personal and Psychological Factors
Institutional and Environmental Factors
Socioeconomic and Family Factors
Health Related Factors
Covid 19 Impact

TA Data Compilation
link
1,166 students 24 courses 4 schools
26,340 observations
Standardized template: School course code & level
Attedance tracked across the 28 sessions
Attedance and participation weightage tracked
Objective triangulation and self reported patterns

Student Self-Report

393 Respondents
4 Schools
30 Factor Items

Section A: Demographics, academic profile, lifestyle, attendance patterns

Section B: Perceptions of absenteeism (Likert, 1–5)
Motivation (High, Moderate, Low)
Importance of Absenteeism
Benefits of Absenteeism

Section C: 30 Likert-scale items across 6 thematic clusters
Section D: Open-ended qualitative feedback
Total scale α = 0.879 - strong internal consistency

Demographics at a Glance

Academic profiles differ by school: SSE and SDSB students have higher CGPAs, SAT scores, and course loads, while MGSHSS and SAHSOL are lower on these metrics. MGSHSS has the highest female representation, SSE the lowest share of financial aid and hostel students, and SAHSOL students are more likely to live on campus and join extracurriculars. SSE also records the highest average absence index.

MGHSS(N = 112)

Variable	Value
CGPA (Mean ± SD)	3.23 ± 0.50
Avg Credit Hours	15.96 ± 1.77
Number of Courses	4.49 ± 0.76
Absence Index	3.09 ± 2.25
Gender	51.8% Female, 48.2% Male
Year of Study	33.0% First, 26.8% Second, 20.5% Third, 18.8% Fourth

SDSB (N = 84)

Variable	Value
CGPA (Mean ± SD)	3.31 ± 0.50
Avg Credit Hours	16.00 ± 0.95
Number of Courses	4.98 ± 0.56
Absence Index	3.85 ± 3.84
Gender	35.7% Female, 64.3% Male
Year of Study	46.4% First, 21.4% Second, 10.7% Third, 21.4% Fourth

SAHSOL (N = 23)

Variable	Value
CGPA (Mean ± SD)	3.10 ± 0.57
Avg Credit Hours	15.87 ± 1.29
Number of Courses	4.52 ± 0.51
Absence Index	3.62 ± 4.24
Gender	34.8% Female, 65.2% Male
Year of Study	30.4% First, 26.1% Second, 13.0% Third, 17.4% Fourth, 13.0% Fifth

SSE (N = 174)

Variable	Value
CGPA (Mean ± SD)	3.35 ± 0.47
Avg Credit Hours	16.24 ± 1.66
Number of Courses	5.05 ± 1.00
Absence Index	5.35 ± 5.48
Gender	29.3% Female, 70.7% Male
Year of Study	34.5% First, 24.7% Second, 19.5% Third, 18.4% Fourth

Perceptions Match Reality: Strong Evidence of True Attendance Differences

“Do students who say they attend regularly actually behave differently from those who say they don’t?”

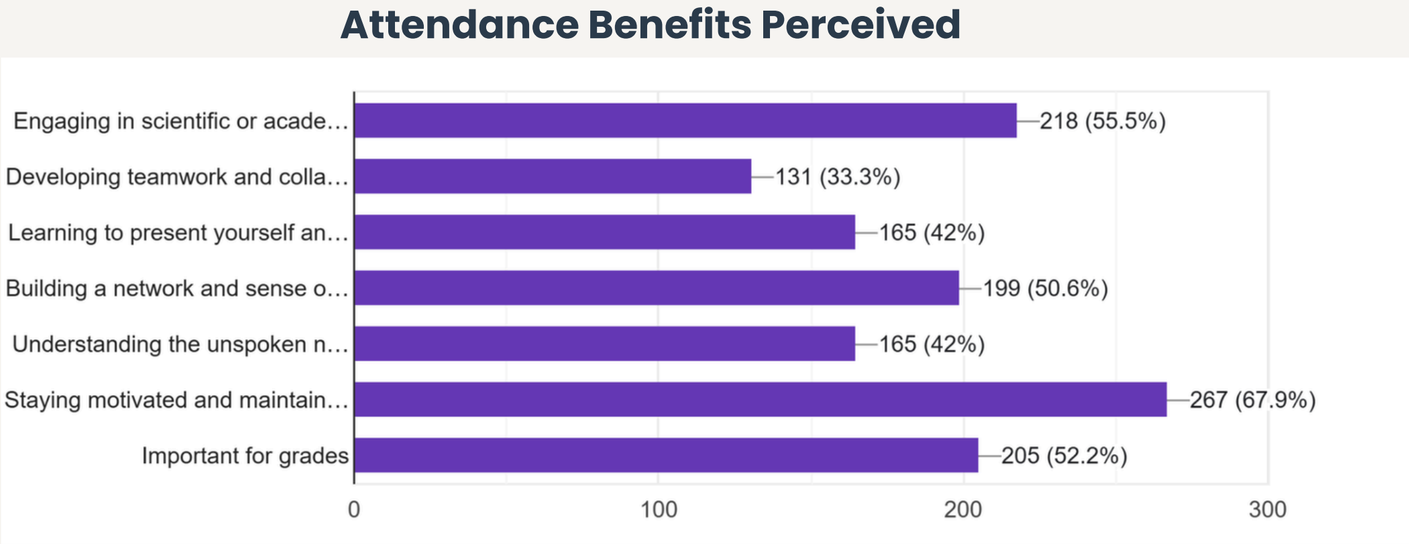
Self-Reported Attendance Patterns vs. Self-Reported Absence (Fall 2025-26)

Self-Reported Pattern	Students (N)	%	Mean Sessions Missed/Course	Multiplier vs. Baseline
Consistently attend	196	49.90%	2.48	1.0× (Baseline)
Attend regularly, miss occasionally	152	38.70%	5.09	2.1×
Frequently miss	42	10.70%	9.46	4.6×
Rarely attend	3	0.80%	~6.5	(Limited data)

- ANOVA test confirms the three self-reported attendance groups differ in actual absence behavior.
- The large F-ratio ($F = 42.37$, $p < 0.001$) indicates that the between-group variance is significantly larger than the within-group variance, meaning the groups are genuinely different from each other, not just randomly varied.

Student Beliefs: Perceived Attendance Benefits

Attendance improves when students see clear academic returns, but social or abstract benefits alone are not strong enough to sustain consistent attendance.



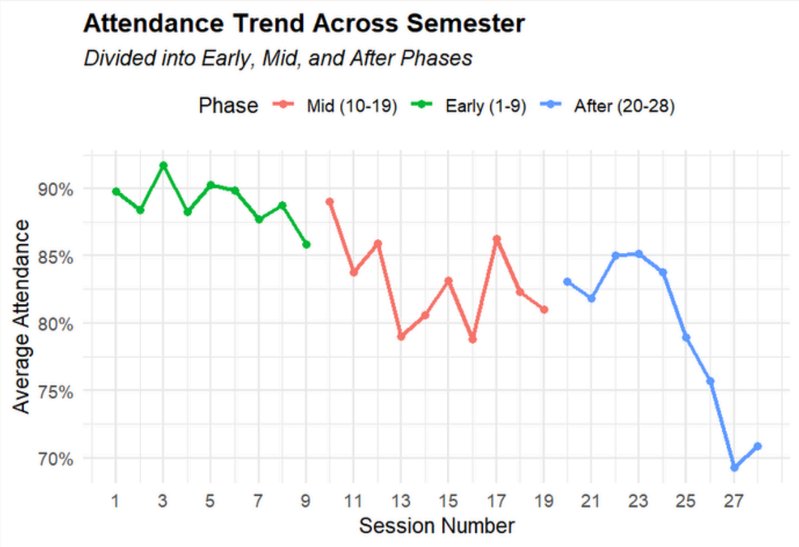
Perceived Benefit Selected	Mean Absence (Selected)	Mean Absence (Not Selected)
Important for grades	3.6	5.02
Learning to present professionally	3.81	4.63
Staying motivated / consistent routine	4.18	4.49

Key Insights

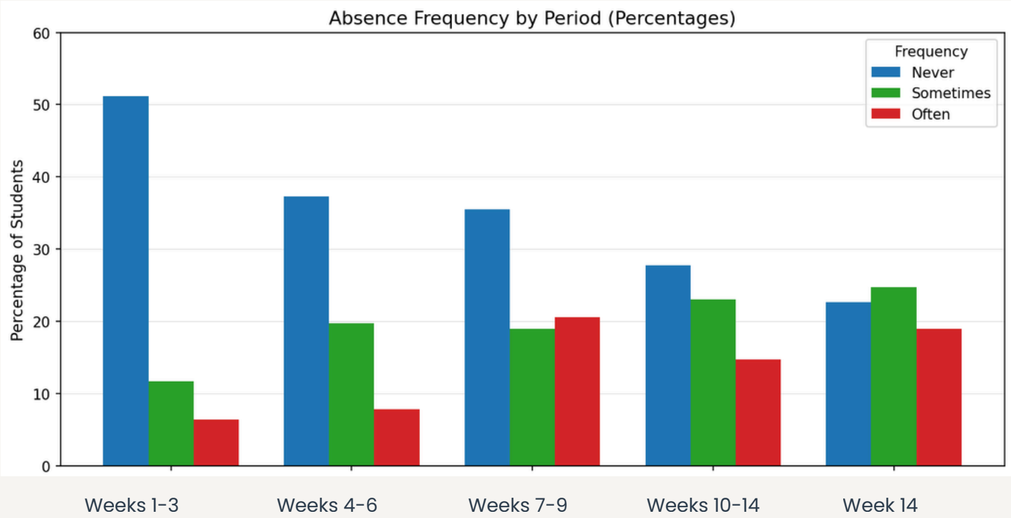
- Students most commonly associate attendance with **staying motivated (67.9%), academic engagement (55.5%),** and **grades (52.2%).**
- **Grade-linked motivation is most effective –** students who see attendance as important for grades miss significantly fewer classes (3.6 vs 5.02).

From Engagement to Absence: Identifying Trends in Attendance

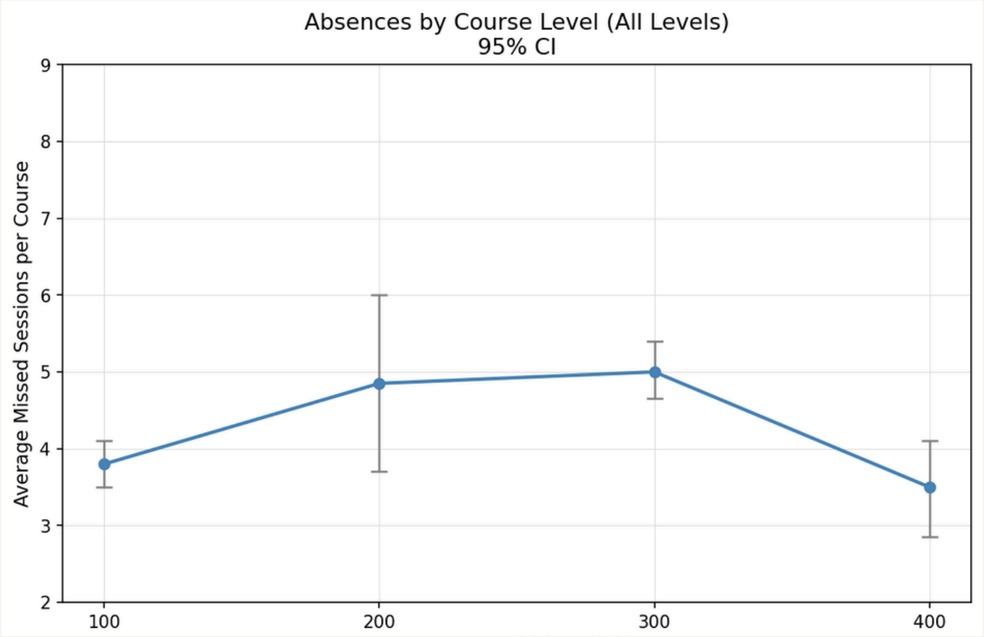
Attendance is systematically linked to the academic calendar, not randomly distributed.



Strong baseline attendance before pressure builds up. Attendance declines mid semester and falls sharply towards the end.



Mapping Absence by Course Levels



Mean Absence by Course Level suggests Low Absence for 100 Level Courses which then increases for 200 to 300 Level Courses

Course Level	Mean Absence	95% CI	n
100-level	3.81	[3.40, 4.22]	963
200-level	4.8	[3.65, 5.95]	462
300-level	4.97	[4.41, 5.52]	316
400-level	3.45	[2.74, 4.17]	122

What students *believe* about the importance of attendance strongly affects how often they actually attend

The more students value attendance, the more they show up.

Belief in Attendance Importance

Belief Level	Students	Mean Sessions Missed/Course	Multiplier
Strongly Agree	35.40%	1.8	Baseline
Agree	29.50%	3	1.7×
Neutral	16.80%	3.1	1.8×
Disagree	12.50%	6	3.4×
Strongly Disagree	5.90%	8.10 (6.3 in pilot)	~3.6×

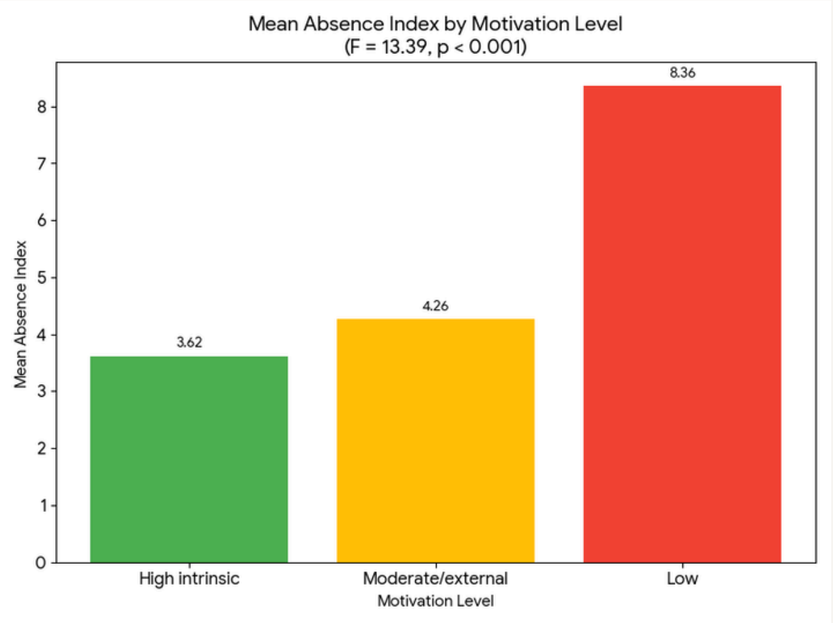
- **Attendance is important. Mean Sessions Missed: 1.8**
- Personal correlation: r = -0.277 (p < 0.001). Students who believe attendance is essential miss significantly fewer classes.
- The mean belief schore is 3.89/5, students generally acknowledge that attendance matters.

Perceived motivation levels for attending classes closely linked to class attendance

Self Reported Motivation Levels

High 39.1% Moderate 54.2 % Low 6.6%

- **High intrinsic motivation:** Mean absence index = 3.62
- **Moderate/external motivation:** mean = 4.26
- **Low motivation:** mean = 8.36
- **ANOVA:** F = 13.39, p < 0.001



Determinants of Absenteeism: Class Timing & Causal Inference, ECON 211 Case Study

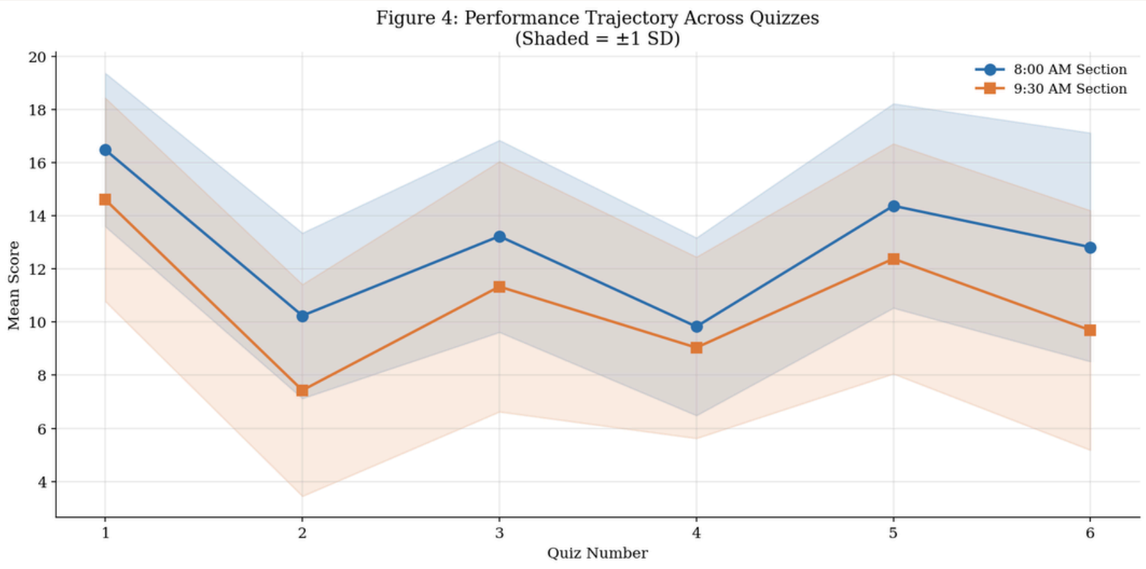
Research question: Does attending an 8:00 AM section (vs. 9:30 AM) causally reduce attendance?

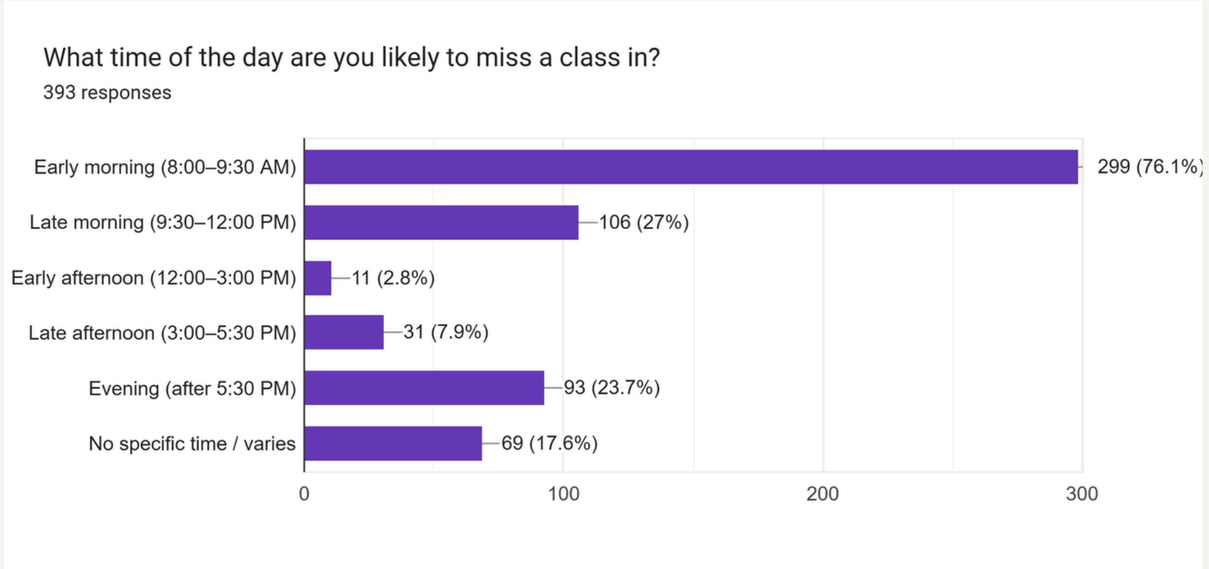
N = 111 students (59 treated: 8 AM; 52 control: 9:30 AM) | Covariate: Quiz 1 score (z-standardised)

Pre-Treatment Imbalance

8:00 AM students scored higher on Quiz 1 (mean z = +0.253 vs. −0.287; Std. Diff = 0.54, far above 0.20 threshold).

Matching is required before causal inference.





Main Finding: The evidence is suggestive but not conclusive , 8:00 AM class timing appears to modestly reduce attendance by approximately 1.3 sessions (out of 25), consistent with the hypothesis that early morning scheduling creates a practical barrier. The effect is not statistically significant at 5%, likely due to small sample size.

Determinants of Absenteeism - Regression Analysis

Category	Factors
Socioeconomic & Logistical	transport_cost, commute_distance, family_responsibilities, work_commitments,
Health	academic_stress, mental_health, physical_health, menstrual_health
Personal & Behavioral	low_motivation, attendance_unnecessary, major_dissatisfaction, leisure_preference, poor_sleep, time_management,
Instructor & Instructional	poor_teaching, unclear_goals, classroom_disruption, overcrowding
Institutional & Policy	attendance_not_graded, online_materials, ai_tools, heavy_workload, participation_fear, class_timing
Social & Peer Dynamics	peer_influence, low_belonging, social_media, extracurriculars

— Academic Controls —					
cgpa	−1.2252	0.474	−2.587	0.010 ★★	[−2.157, −0.294]
year_numeric	0.0872	0.205	0.426	0.67	[−0.316, 0.490]
— School Fixed Effects (ref: MGHSS) —					
school_SAHSOL	0.6245	1.067	0.585	0.559	[−1.474, 2.723]
school_SDSB	0.8881	0.621	1.43	0.154	[−0.333, 2.110]
school_SSE	2.3312	0.564	4.137	0.000 ★★★	[1.223, 3.439]
— Thematic Composites (6 Categories) —					
Socioeconomic & Logistical	−0.4865	0.274	−1.773	0.077 †	[−1.026, 0.053]
Health	−0.2763	0.256	−1.080	0.281	[−0.779, 0.227]
Personal & Behavioral	0.8697	0.394	2.209	0.028 ★★	[0.095, 1.644]
Instructor & Instructional	−0.1646	0.236	−0.698	0.486	[−0.629, 0.299]
Institutional & Policy	0.5151	0.304	1.697	0.091 †	[−0.082, 1.112]
Social & Peer Dynamics	0.3957	0.338	1.171	0.242	[−0.269, 1.060]

- CGPA has a significant negative effect (p = 0.010), meaning higher CGPA is associated with a lower outcome variable.
- SSE students differ significantly from MGHSS (large positive effect, p < 0.001), while other schools are not significant.
- Among thematic factors, Personal & Behavioral is positively significant (p = 0.028).
- Socioeconomic & Logistical and Institutional & Policy are marginally significant (10%), suggesting weaker evidence of an effect.

Determinants of Absenteeism - Regression Analysis

Personal Behavior and Motivation opened up

Variable	Coef	Std Err	t	p-value	95% CI Lower	95% CI Upper	Sig
CGPA	−1.195	0.478	−2.500	0.013	−2.135	−0.255	**
School: SSE	2.258	0.563	4.012	0	1.151	3.365	***
c11: Attendance Unnecessary	0.452	0.209	2.158	0.032	0.04	0.863	**
c12: Major Dissatisfacti on	0.428	0.208	2.058	0.04	0.019	0.838	**
c16: Other Challenges	−1.070	0.311	−3.445	0.001	−1.680	−0.459	***
c15: Weak Time Managemen	0.364	0.221	1.643	0.101	−0.072	0.799	†
c10: Low Motivation	−0.079	0.216	−0.366	0.715	−0.503	0.345	—
c13: Prefer Leisure	−0.005	0.203	−0.024	0.981	−0.405	0.395	—
c14: Poor Sleep	0.148	0.213	0.695	0.487	−0.271	0.567	—

- CGPA and personal challenges matter most: Higher CGPA and more personal challenges reduce the likelihood of the outcome.
- School differences matter: Students from SSE show a significantly higher likelihood compared to the reference school.
- Believing that attendance is unnecessary significantly increases absenteeism (β = 0.452, p = 0.032), suggesting mindset plays an important role.

Determinants of Absenteeism - Regression Analysis

Institutional Policy opened up

actor	Coefficient	p-value	Direction	Significance
School: SSE	2.336	< 0.001	↑ More absences	1% ***
CGPA	−1.289	0.007	↓ Fewer absences	1% ***
Personal & Behavioral factors	0.911	0.023	↑ More absences	5% **
Attendance not graded (C21)	0.31	0.068	↑ More absences	10% *
Socioeconomic & Logistical factors	−0.482	0.081	↓ Fewer absences	10% *

Question:
To what extent do the following factors influence you to miss a class?
Attendance does not count towards my overall grade.

- At 10% (marginally significant): When attendance isn't graded (C21), absences tend to be higher — pointing to accountability as a deterrent.
- Socioeconomic & Logistical factors slightly reduce absences, though this is only borderline significant.

Impact of Absenteeism - Regression Analysis

Variable	Reg 1A – Linear			
	Coef	Std Err	z	p-value
Missed sessions	−0.033	0.017	−1.971	0.049
Missed sessions² (quadratic)	—	—	—	—
SAT Score	0.232	0.098	2.376	0.018
Financial Aid (Yes)	−0.423	0.22	−1.920	0.055
EDU (course type)	1.537	0.401	3.831	0
HSS (course type)	2.879	0.418	6.895	0
LAW (course type)	1.056	0.575	1.838	0.066
POL (course type)	0.811	0.353	2.295	0.022
OTHER (course type)	0.723	0.297	2.436	0.015
BIO (course type)	−1.215	0.645	−1.885	0.059
CHEM (course type)	0.537	0.308	1.745	0.081
CS (course type)	—	—	—	—
SOC (course type)	—	—	—	—

- Coefficient on missed sessions = -0.0333
- Each additional missed session → -0.0333 change in grade (ordinal units)
- ~30.0 missed sessions ≈ 1 full grade drop
- 10 missed sessions → -0.333 grade change

F → 1 | D → 2 | C- → 3 | C → 4| C+ → 5 | B- → 6 | B → 7 | B+→ 8 | A- → 9 | A → 10 | A+ → 11

Key Takeaways

- Drop in attendance mid-semester followed by a continued decline towards finals.
  - At-risk Periods: Attendance declines during high-pressure phases (mid-terms, finals, major deadlines), when competing demands intensify.

- Absenteeism is driven by:
  - Personal and behavioral factors
    - Dissatisfaction with major
    - Attendance Unnecessary
    - Time management
  - School-level variation based on internal policy:
    - SSE has high absence rates.
  - Class Timing
    - Caveat: Higher performing students self-select into morning slots.
    - Implication: Students should not be force enrolled into early morning sections.
  - Juniors are least likely to attend classes.

- Impact on Grades:
  - Evidence suggests a positive but marginal effect - e.g., improvements from B to B+ if a student attends 30 classes.
  - Attendance supports performance, but is not the sole driver; other factors (ability, effort) also matter.

Why emphasize attendance:

- Attendance is encouraged as a proxy for engagement, discipline, and continuous learning.
  - If the institutional aim is to incentivize attendance, target interventions mid-semester through finals, when drop-offs are most likely.